

Ali Naqvi

+1 437-329-9585 | anaqvee02@gmail.com | [GitHub](#)

EDUCATION

University of Waterloo
Bachelor of Mathematics (BMath)

Waterloo, ON
Expected April 2031

TECHNICAL SKILLS

Languages: Python, Rust, C, C++, JavaScript, TypeScript, HTML, CSS

Technologies: GitHub, PyTorch, Hugging Face, FastAPI, Docker, TensorFlow, Keras, Flask, React

EXPERIENCE

EPSi, Florida International University

July 2026 – August 2026

Undergraduate Research Intern

Miami, FL (Remote)

- * Engineered a privacy-preserving Bonawitz-style **Secure Federated Learning** pipeline using a **PyTorch 1D-CNN** (46k parameters) to detect solar inverter faults across 37 clients, achieving **85.2% accuracy** and an **88.3% F1 score** over 20 training rounds.
- * Implemented a 4-round cryptographic protocol leveraging **ECDH P-384**, Shamir's Secret Sharing over **$\text{GF}(2^{521} - 1)$** , double masking, and **ECDSA signatures**, successfully reducing gradient leakage similarity from 0.98 to **0.04** with **0.0 error** in aggregate reconstruction.
- * Optimized system scalability and resilience to cleanly withstand up to **30% client dropout** without aggregate corruption, while profiling edge-device feasibility to run secure aggregation in **3.84s** with single-core proxy inference under **0.22ms**.

St. Joseph's Healthcare Hamilton

June 2026 – Current

Machine Learning Intern

Hamilton, ON

- * Engineered a clinical-grade NLP routing engine by fine-tuning **Llama-3** on matched triads to understand complex medical semantics, achieving **99.9% accuracy** with a **100% sensitivity** oncology failsafe.
- * Conquered syntactic traps that bypass standard regex pipelines, reducing false clinical alarms by over **80%** by training the model to distinguish contextual nodule statuses and locations.

PROJECTS

mtdi | Rust, Systems Programming, arm64 Assembly

- * Built and open-sourced a high-performance **dynamic instrumentation engine** for arm64 macOS, achieving tracing speeds **10–100x faster than Frida** and matching Dobby.
- * Engineered low-level inline hook detours providing **1.6 ns/call** syscall execution and **15 ns/call** full-context hooks, featuring live-PID injection via **task_for_pid** and remote thread bootstrapping.
- * Designed a custom probe compiler with an **AST verifier** enforcing zero-panic rules, integrated with an allocation-free **SPSC ring buffer** logging pipeline to ensure zero-blocking, lock-free telemetry.

PicoEmulator | Python, Cython, ARM Architecture

- * Built a high-performance **ARM Cortex-M0+ emulator** in **Python** that simulates the Raspberry Pi Pico's RP2040 microcontroller, implementing the full **Thumb-1 instruction set**.
- * Achieved a **1000x speedup** over pure Python by engineering the core in **Cython**, utilizing **direct memory access** and inline instruction fast-paths to run at ~ 33 MHz.
- * Integrated comprehensive **benchmarking tools** to rigorously measure performance across various technical workloads.